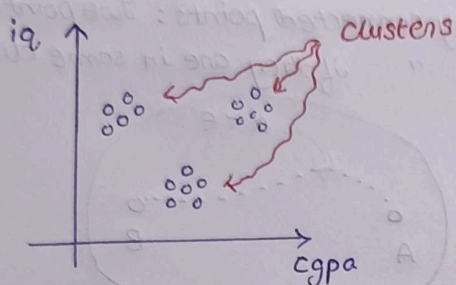


Unsupervised Learning - Clustering

K-Means Clustering : (Intuition)

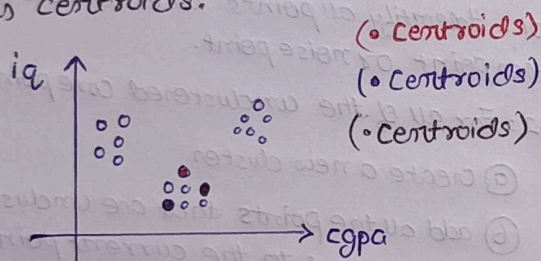
=> Basically it helps in forming clusters.



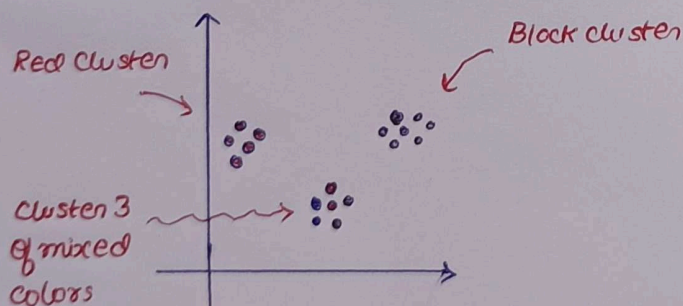
=> But how it works?

① Decide n clusters (say $K=3$).

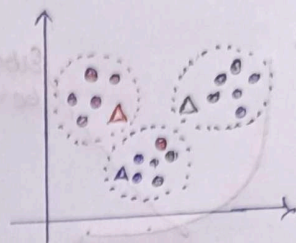
② For value of K it takes any K random points as centroids.



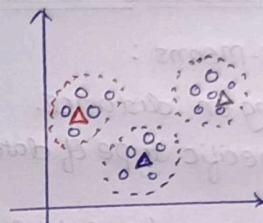
③ Now, for each point on the data, the euclidean distance is calculated from each centroid. The nearest centroid color is assigned to that point as its cluster colour. Thus, each point is assigned to a cluster.



④ Move the ~~clusters~~ centroids. For this we find the centroid of each cluster.



⑤ we will keep doing this each cluster has same color and centroid stop moving.

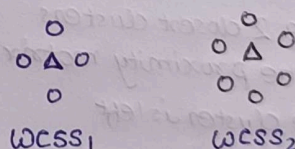


=> But how to decide no. of K ? we use Elbow method. This plots graph b/w No. of clusters and WCSS (within cluster sum of squared distance) or inertia.

But how to calculate WCSS of any cluster? It's the sum of square of distance of centroid of cluster from its each point.

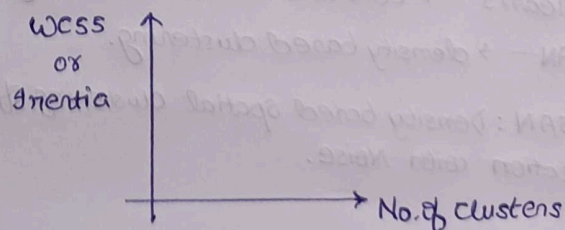
$$WCSS = d_1^2 + d_2^2 + d_3^2 + \dots$$

For multiple clusters, it be sum of each clusters WCSS.

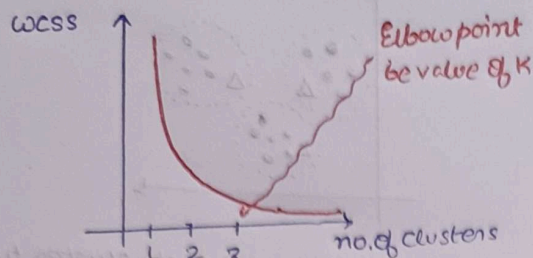


$$\therefore WCSS \text{ of them} = WCSS_1 + WCSS_2$$

Btw, what be on X and Y axis?



The graph look like:



Agglomerative Hierarchical Clustering:

⇒ Drawbacks of K-means:

- ① Too much relying on distance.
- ② works on a specific type of data set only.

⇒ To overcome this we have two more clustering methods namely:

- (i) Hierarchical clustering
- (ii) Density Based clustering

⇒ Types of Hierarchical clustering:

- ① Agglomerative clustering
- ② Divisive clustering

⇒ Agglomerative clustering: (algorithm used)

- ① Initialise the Proximity matrix
- ② Make each point a cluster
- ③ Inside a loop
 - a. Merge the 2 closest clusters
 - b. update the proximity matrix
- ④ until only one cluster is left

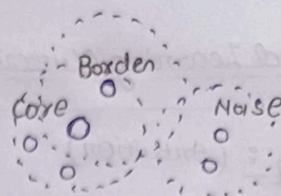
DBSCAN Clustering:

⇒ K-means → centroid based clustering.

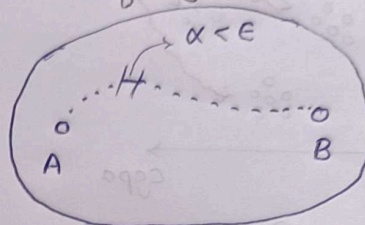
DBSCAN → density based clustering.

⇒ DBSCAN: Density based spatial clustering of Application with Noise.

⇒



⇒ Density connected points: Two points are called density " if they are in same cluster.



⇒ Algorithm:

Step 0: minpts / epsilon

Step 1: Identify all points as either core point, border point or noise point.

Step 2: For all of the unclustered core points:

① Create a new cluster

② add all the points that are unclustered and densely connected to the current point into this cluster.

Step 3: For each unclustered border point assign it to the cluster of nearest core point.

Step 4: leave all the noise points as it is.

